Case (ii)

Consider the system of linear equations

$$3x+2y = 5, \quad \text{... (1)}$$

$$6x+4y = 10 \quad \text{... (2)}$$

Using equation (1), we get

$$x = \frac{5-2y}{3} \quad \text{... (3)}$$

Substituting (3) in (2) and simplifying, we get $0 = 0$.

This informs us that equation (2) is an elementary transformation of equation (1). In fact, by dividing equation (2) by 2, we get equation (1). It is not possible to find uniquely x and y with just a single equation (1).

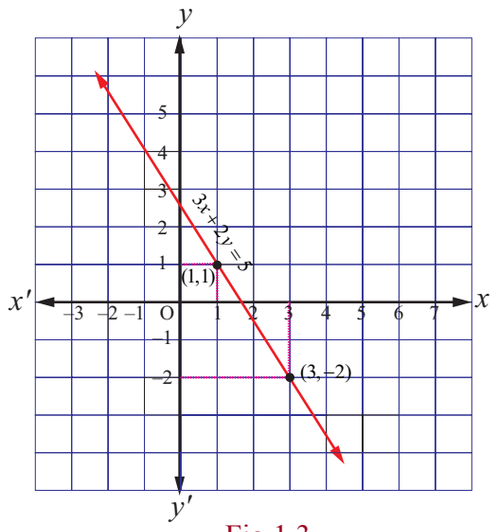


Fig.1.3

So we are forced to assume the value of one unknown, say $y = t$, where t is any real number. Then, $x = \frac{5-2t}{3}$. The two equations (1) and (2) represent one and only one straight line (coincident lines) in two dimensional analytical geometry (see Fig. 1.3) . In other words, the system is consistent (a solution of (1) is also a solution of (2)) and has infinitely many solutions, as t can assume any real value.

Case (iii)

Consider the system of linear equations

$$4x+y = 6, \quad \text{... (1)}$$

$$8x+2y = 18. \quad \text{... (2)}$$

Using equation (1), we get

$$x = \frac{6-y}{4} \quad \text{... (3)}$$

Substituting (3) in (2) and simplifying, we get $12 = 18$.

This is a contradicting result, which informs us that equation (2) is inconsistent with equation (1). So, a solution of (1) is not a solution of (2).

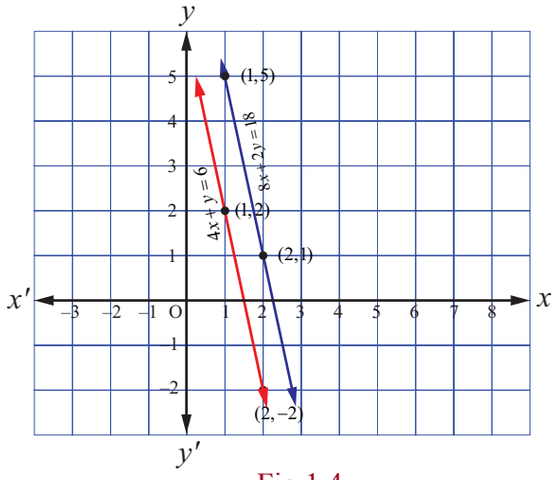


Fig.1.4

In other words, the system is inconsistent and has no solution. We note that the two equations represent two parallel straight lines (non-coincident) in two dimensional analytical geometry (see Fig. 1.4). We know that two non-coincident parallel lines never meet in real points.

Note

(1) Interchanging any two equations of a system of linear equations does not alter the solution of the system.

(2) Replacing an equation of a system of linear equations by a non-zero constant multiple of itself does not alter the solution of the system.

(3) Replacing an equation of a system of linear equations by addition of itself with a non-zero multiple of any other equation of the system does not alter the solution of the system.

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